Sapienza Università di Roma

Department of Mechanical and Aerospace Engineering

Homework 3 — Attitude Control of a Launch Vehicle in Atmospheric Flight

Dynamics and Control of Launch Vehicles

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Contents

1 Introduction

This report addresses the design of an attitude control system for a launch vehicle (LV) during the atmospheric phase of flight, at the instant of maximum dynamic pressure ($t = 72$ s, max-$\bar{q}$). This is the hardest point to control: the aerodynamic moment destabilises the airframe and the wind disturbance is largest. I design the controller with classical frequency-domain tools, using the Nichols chart as the main stability indicator, following the short-period pitch-plane model of Greensite [1].

1.1 Design methodology: frozen coefficients at max-$\bar{q}$

Launch-vehicle attitude control in atmospheric flight is classically handled by the **time-slice** (frozen-coefficient) method: the slowly varying mass, inertia and aerodynamic coefficients are frozen at a representative instant, a linear time-invariant short-period model is built and analysed at that operating point, and the controller is then validated against time-varying simulations [1, Sec. 2]. The main sizing disturbance is the wind profile, which drives both the trajectory dispersion and the structural bending loads—hence the gust analysis of Tasks 1 and 2. I pick $t = 72$ s because it is the load- and stability-critical slice: the aerodynamic destabilising moment and the wind-induced angle of attack are both near their maxima. The catch is that point-wise Nichols margins certify only the frozen model and do not by themselves guarantee stability of the underlying time-varying system; this is what motivates the parametric robustness study of Task 3 and, eventually, gain scheduling along the trajectory.

1.2 Plant model

The pitch-plane short-period dynamics are described by the linear time-invariant model (assignment Eq. 1), with state vector $\mathbf{x} = [\, z,\ \dot{z},\ \theta,\ \dot{\theta},\ \eta,\ \dot{\eta}\,]^\top$ (lateral drift, pitch attitude and first bending mode), control input δ (TVC deflection) and disturbance α_w (wind angle of attack):

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & a_1 & a_1 V + a_4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & A_6/V & A_6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -\omega_1^2 & -2\zeta_1\omega_1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ a_3 \\ 0 \\ K_1 \\ 0 \\ -\phi_{1,TVC} T_c \end{bmatrix} \delta + \begin{bmatrix} 0 \\ -a_1 V \\ 0 \\ -A_6 \\ 0 \\ 0 \end{bmatrix} \alpha_w. \quad (1)$$

The pitch channel is the core of the problem and follows from a moment balance about the centre of mass. The aerodynamic moment $N_\alpha l_\alpha$ acts on the local incidence—built from the pitch attitude θ, the plunging term $\dot{z}/V$ and the wind angle α_w—while the gimballed thrust contributes a control moment $T_c l_c\,\delta$; dividing the balance $I_{yy}\ddot{\theta} = N_\alpha l_\alpha(\theta + \dot{z}/V - \alpha_w) + T_c l_c\,\delta$ by I_{yy} reproduces row 4 of Eq. (1),

$$\ddot{\theta} = A_6\,\theta + \frac{A_6}{V}\,\dot{z} + K_1\,\delta - A_6\,\alpha_w, \tag{2}$$

and identifies the two structural coefficients: the **aerodynamic moment** $A_6 = N_\alpha l_\alpha / I_{yy} > 0$ (denoted μ_α) and the **control effectiveness** $K_1 = T_c l_c / I_{yy}$ (μ_c). With $A_6 > 0$ the airframe transfer function $\theta/\delta = K_1/(s^2 - A_6)$ carries an open-loop pole at $s = +\sqrt{A_6} = +1.84$ rad/s in the right half-plane: the **statically unstable** airframe of a vehicle whose centre of pressure lies ahead of its centre of mass, for which feedback is mandatory [2, Lecture 15]. The TVC coefficient K_1 sets the control authority available to counter it.

The sensed quantities are produced by the inertial unit (assignment Eq. 2), which couples the bending coordinate η into the gyro and accelerometer channels through $\sigma_{1,ins}$ and $\phi_{1,ins}$:

$$\theta_m = \theta + \sigma_{1,ins}\,\eta, \quad \dot{\theta}_m = \dot{\theta} + \sigma_{1,ins}\,\dot{\eta}, \quad z_m = z - \phi_{1,ins}\,\eta, \quad \dot{z}_m = \dot{z} - \phi_{1,ins}\,\dot{\eta}. \tag{3}$$

This contamination is exactly what makes the lightly damped first bending mode—written with $\omega_1 \equiv \omega_{BM} = 18.9$ rad/s and $\zeta_1 \equiv \zeta_{BM} = 0.005$ in Eq. (1)—visible to the controller and drives the need for a dedicated filter (Task 2).

The actuator is the second-order TVC of Eq. 3,

$$W_{TVC}(s) = \frac{\omega_{TVC}^2}{s^2 + 2\zeta_{TVC}\omega_{TVC}s + \omega_{TVC}^2}, \qquad \omega_{TVC} = 70 \text{ rad/s},\ \zeta_{TVC} = 0.7, \tag{4}$$

in series with a pure transport delay $\tau = 20$ ms. The delay is approximated by a third-order Padé rational function $e^{-\tau s} \approx P_3(s)$, the diagonal [3/3] member of the family obtained by matching the Taylor expansion of $D(s)\,e^{-\tau s} = N(s)$ to the highest possible order; the illustrative second-order form is

$$e^{-\tau s} \approx \frac{12 - 6\,\tau s + (\tau s)^2}{12 + 6\,\tau s + (\tau s)^2}, \tag{5}$$

an **all-pass** rational function ($|P(j\omega)| = 1$ for all ω) that reproduces the phase lag of the delay while leaving the loop gain unchanged. I keep order three, matching the implementation, so that the approximation stays accurate up to the bending frequency, where $\omega_{BM}\tau \approx 0.38$ is no longer negligible. Parameters at $t = 72$ s are read from the reference data set `GreensiteLPV_DATA.mat` (cross-checked against Table 1 of the assignment); the salient values are reported in Table 1.

Table 1: Pitch-plane LV parameters at the max-$\bar{q}$ condition ($t = 72$ s).

Quantity	Symbol	Value
Aerodynamic moment	$A_6 = \mu_\alpha$	3.382 s^{-2}
Control effectiveness	$K_1 = \mu_c$	4.565 s^{-2}
Drift coefficients	a_1, a_3, a_4	-0.0154, 20.61, -27.27 s^{-2}
Relative velocity	V	937.7 m/s
Bending mode	ω_{BM}, ζ_{BM}	18.9 rad/s, 0.005
INS coupling	$\phi_{1,ins}$ [–], $\sigma_{1,ins}$ [rad/m]	0.8, 0.178
TVC actuator	$\omega_{TVC}, \zeta_{TVC}, \tau$	70 rad/s, 0.7, 20 ms

1.3 Control architecture

A proportional–derivative attitude law is adopted, augmented by a weak feedback on the lateral-drift channel:

$$\delta_{cmd} = K_{P,\theta}\,(\theta_{ref} - \theta_m) - K_{D,\theta}\,\dot{\theta}_m - K_{P,z}\,z_m - K_{D,z}\,\dot{z}_m, \tag{6}$$

with $K_{P,z}, K_{D,z}$ small and negative, of order 10^{-3}, as recommended in the assignment guidelines. Closing the loop also on $(z_m, \dot{z}_m)$ trades between two classical objectives in wind: containing the lateral drift (drift minimum) and containing the angle of attack $\alpha = \theta + \dot{z}/V + \alpha_w$, hence the structural load $\bar{q}\,\alpha$ (load minimum) [2, Lecture 17] — the binding concern at the max-$\bar{q}$ condition studied here. The weak negative gains keep the drift bounded without fighting the natural load-relieving weathercock of the airframe; I therefore monitor both z and $\bar{q}\,\alpha$ in the gust simulations. The actual deflection follows from the actuator and the bending filter $H_x(s)$ (Task 2): $\delta = W_{TVC}(s)\,e^{-\tau s}\,H_x(s)\,\delta_{cmd}$. The complete loop is sketched in Figure 1.

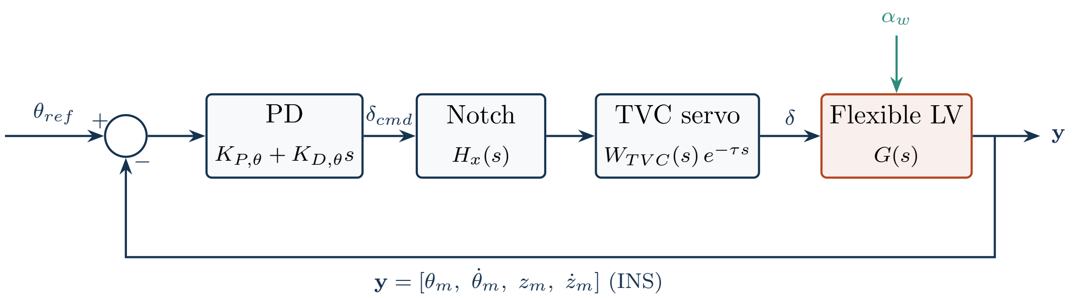

Figure 1: Full attitude-control loop (Tasks 2–3): the PD law, the bending notch H_x, the TVC servo with transport delay, and the flexible airframe forced by the wind angle of attack α_w. The INS measurement vector $\mathbf{y}$ closes the loop; its lateral-drift channels carry the weak $K_{P,z}, K_{D,z}$ feedback of Eq. (6). Task 1 uses the same loop with $H_x = 1$, $W_{TVC} = 1$ and the rigid plant (Figure 4).

1.4 A note on conditional stability

Because the open-loop airframe carries a single right-half-plane pole (the $+1.84$ rad/s aerodynamic pole $s = +\sqrt{\mu_\alpha}$), the loop is **conditionally stable**: the proportional gain must be large enough to overcome the static aerodynamic instability, $K_{P,\theta} > \mu_\alpha/\mu_c$, so stability is lost if the loop gain is reduced below this threshold, not only if it is increased [2, Lecture 15]. An aerodynamically unstable vehicle driven through a second-order (or higher) actuator therefore has two gain margins—a low-frequency aerodynamic margin (reduce gain → unstable) and a high-frequency rigid-body margin (increase gain → unstable)—which on the Nichols chart show up as two 0 dB crossings of the $-180°$ line [2, Lecture 16]. The margins are therefore read on the *full* open loop and classified by frequency band, rather than taken from `margin`'s default crossover: the aerodynamic gain margin at the low-frequency $-180°$ crossing (a gain-reduction margin, $|GM_a| \approx 6$ dB), the rigid-body phase margin at the gain crossover ($PM \approx 30°$), and—once the actuator and bending are present (Task 2)—the high-frequency rigid-body gain margin and the bending Flex margins. This is standard practice for the attitude control of aerodynamically unstable launchers [3].

The same conditional stability is visible in the s-plane: the root locus of the rigid loop (Figure 2) starts at the open-loop poles, one of which — the aerodynamic pole $+\sqrt{\mu_\alpha}$ — lies in the right half-plane, and, as the loop gain grows, that pole is drawn across the imaginary axis into the left half-plane. Stabilising the airframe therefore demands a *minimum* loop gain, $K_{P,\theta} > \mu_\alpha/\mu_c$; the upper bound that closes the finite stable band comes from the actuator lag and is read on the Nichols chart.

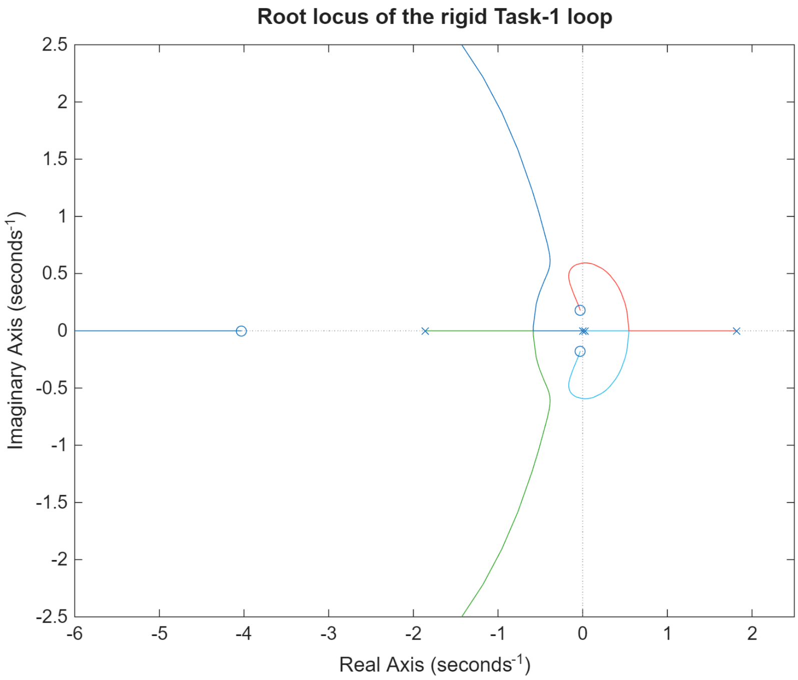

Figure 2: Conditional stability on the root locus of the rigid Task-1 loop (built-in `rlocus`). Following the usual convention, $\times$ marks the open-loop poles (where the branches start, $k = 0$) and $\circ$ the open-loop zeros (where they end, $k \to \infty$); the branches trace the closed-loop poles as the loop gain grows. As the loop gain grows the unstable aerodynamic pole at $+\sqrt{\mu_\alpha} = +1.84$ rad/s ($\times$ in the right half-plane) is drawn into the left half-plane, so the closed loop is stable at the design gain; the drift mode leaves the real axis into the lightly damped pair near the imaginary axis. Stability therefore requires a *minimum* loop gain, $K_{P,\theta} > \mu_\alpha/\mu_c = 0.74$ (the low-frequency aerodynamic margin); the high-frequency margin set by the actuator lag is the second 0 dB crossing on the Nichols chart.

2 Task 1 — Rigid Launch Vehicle

Figure 3 sketches the design pipeline of `main_task1.m`, which Tasks 2 and 3 reuse with richer plant and actuator models.

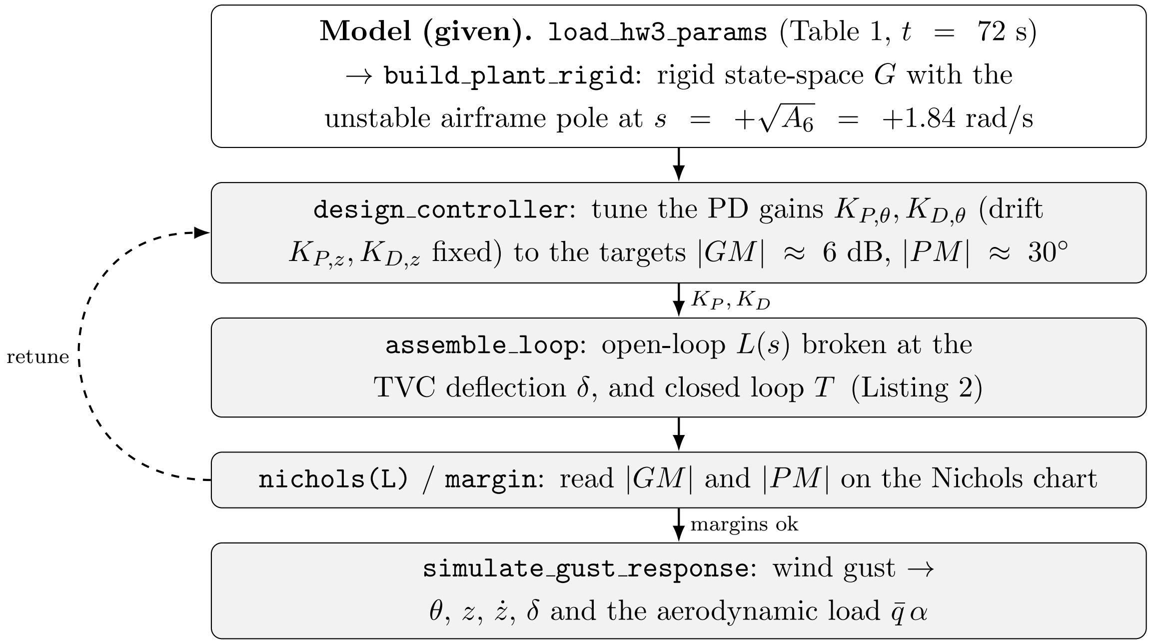

Figure 3: Task 1 design pipeline (`main_task1.m`). Tasks 2 and 3 reuse it with the full flexible plant and the TVC+delay+notch chain `Wact`. Listing numbers refer to Appendix A.

2.1 Rigid model and loop

Neglecting the bending mode, the plant reduces to the four states $[z, \dot{z}, \theta, \dot{\theta}]$ (`build_plant_rigid.m`, Listing 1) and the actuator is taken as ideal: $W_{TVC}(s) = 1$ and no transport delay. The open-loop transfer broken at the TVC deflection, in the $1 + L$ convention, is obtained by closing the static PD law of Eq. (6) around the plant (`assemble_loop`, Listing 2):

$$L(s) = K_{P,\theta}\, G_{\delta\to\theta}(s) + K_{D,\theta}\, G_{\delta\to\dot{\theta}}(s) + K_{P,z}\, G_{\delta\to z}(s) + K_{D,z}\, G_{\delta\to\dot{z}}(s), \tag{7}$$

where $G_{\delta\to\bullet}$ are the plant channels from δ to each measurement. The dominant term is the pitch loop $L_\theta(s) = (K_{P,\theta} + K_{D,\theta}s)\,K_1/(s^2 - A_6)$, whose closed-loop characteristic polynomial $s^2 + K_1 K_{D,\theta}\, s + (K_1 K_{P,\theta} - A_6)$ is stable only for $K_{P,\theta} > A_6/K_1 = 0.74$ and $K_{D,\theta} > 0$, so the proportional gain has to overcome the aerodynamic instability. The rigid loop is sketched in Figure 4.

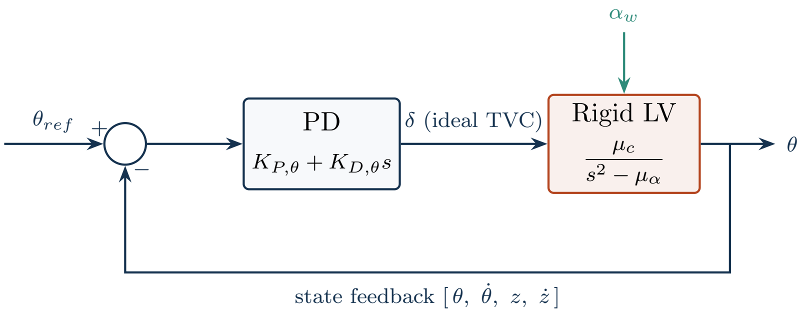

Figure 4: Task 1 rigid loop: the PD law drives the airframe through an ideal actuator ($W_{TVC} = 1$, no delay or notch). The pitch channel shown here, $\mu_c/(s^2 - \mu_\alpha)$, carries the unstable pole at $+\sqrt{\mu_\alpha}$; the full loop with the actuator chain is Figure 1.

The gain and phase margins quoted below are read on the *loop transfer* $L(s)$, obtained by **opening this loop at the TVC deflection** δ (Figure 5): δ is the single actuator signal, so cutting the wire there, injecting a test signal δ_{in} into the plant and reading the returning δ_{out} at the actuator output gives a scalar $L(s) = \delta_{out}/\delta_{in}$. This is why the Nichols charts are labelled "from δ to δ": input and output sit at the same break point. Closed-loop poles are the roots of $1 + L = 0$.

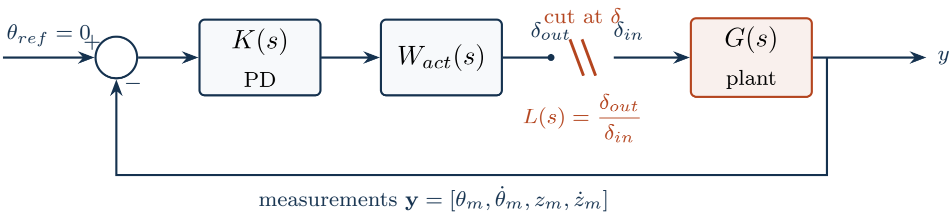

Figure 5: Where the loop is opened. The loop transfer $L(s)$ used for the Nichols charts and the gain/phase margins is formed by cutting the loop at the actuator output δ (the plant input): a test δ_{in} is injected into the plant and the returning δ_{out} is read back around the loop, so $L(s) = \delta_{out}/\delta_{in}$ is a single scalar transfer "from δ to δ". Here $W_{act} = 1$ (ideal actuator, Task 1); for Tasks 2–3 $W_{act} = H_x(s)\, W_{TVC}(s)\, e^{-\tau s}$ (notch, TVC servo and delay).

2.2 Tuning

The pitch gains follow the classical launcher design of the course notes [2, Lecture 16]. Starting from the canonical closed form on the decoupled rotational dynamics $\ddot{\theta} = A_6\theta + K_1\delta$,

$$K_{P,\theta}^{(0)} = \frac{2A_6}{K_1}, \qquad K_{D,\theta}^{(0)} = \frac{\sqrt{A_6}}{K_1}, \tag{8}$$

which place the pitch pair at $\omega_c = \sqrt{A_6}$, $\zeta = 0.5$ and give a 6 dB gain margin and 30° phase margin *on that decoupled loop*. On the full four-state loop the lateral-drift feedback erodes the aerodynamic gain margin (the 6 dB drops to about 4 dB), so a two-parameter `fminsearch` retunes $(K_{P,\theta}, K_{D,\theta})$ on the *full* loop until the frequency-classified aerodynamic gain margin and rigid-body phase margin meet the assignment targets. The lateral gains stay fixed at $K_{P,z} = K_{D,z} = -10^{-3}$ (load relief, not a stability margin). The result is

$$K_{P,\theta} = 1.78, \quad K_{D,\theta} = 0.44, \quad K_{P,z} = K_{D,z} = -10^{-3}, \tag{9}$$

with, read on the full loop, an aerodynamic gain margin $|GM_a| = 6.0$ dB at 0.59 rad/s, a rigid-body phase margin $PM = 30°$ at 2.45 rad/s, and a delay margin of 213 ms. In the pole-placement language of the course notes [2, Lecture 15] these gains place the pitch pair at $\omega_c = \sqrt{K_1 K_{P,\theta} - A_6} = 2.18$ rad/s — inside the usual 1–4 rad/s control band — with damping $\zeta = K_1 K_{D,\theta}/(2\omega_c) = 0.46$, in the standard 0.4–0.7 range. These relations invert the closed-loop characteristic polynomial, matching $s^2 + K_1 K_{D,\theta}\, s + (K_1 K_{P,\theta} - A_6)$ to $s^2 + 2\zeta\omega_c s + \omega_c^2$:

$$K_{P,\theta} = \frac{\omega_c^2 + A_6}{K_1}, \qquad K_{D,\theta} = \frac{2\zeta\omega_c}{K_1}. \tag{10}$$

Reading the margins. The margins are read on the *full* open loop and classified by frequency band, not taken from `margin`'s default crossover [3]. Because the lateral-drift position feedback carries a free integrator ($z/\delta \to \infty$ at DC), the Nichols curve comes from the top (Figure 6): the aerodynamic gain margin is the low-frequency gain-reduction margin, read where the curve crosses the critical point (+180°) at $|GM_a| = 6$ dB, and the rigid-body phase margin is read at the 2.45 rad/s gain crossover. With the ideal Task-1 actuator these are the only rigid-body margins; the high-frequency rigid-body gain margin (a gain-increase margin set by the actuator lag) and the bending Flex margins appear once the TVC servo, transport delay and notch are added in Task 2. Crossings below ~ 0.5 rad/s are lateral-drift artifacts, not rigid-body margins.

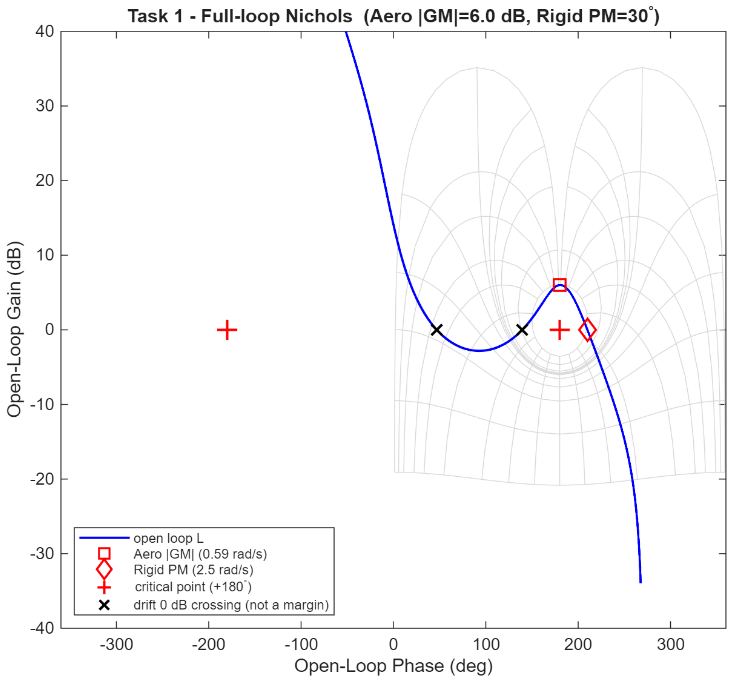

Figure 6: Task 1 — full-loop Nichols in the launch-vehicle convention (critical point at $+180°$). The loop comes from the top (lateral-drift integrator); the design meets $|GM_a| \approx 6$ dB (aerodynamic, at $+180°$) and $PM \approx 30°$ (rigid-body, at the 2.45 rad/s crossover).

In the s-plane the same design reads as a pole placement (Figure 7): the PD gains pull the unstable $+1.84$ rad/s aerodynamic pole into the left half-plane, leaving a well-damped fast pitch pair ($-0.95 \pm 1.90i$, $\zeta \approx 0.45$) and a slow, lightly damped drift pair ($-0.06 \pm 0.23i$) close to the imaginary axis — the stability margin the Nichols chart quantifies.

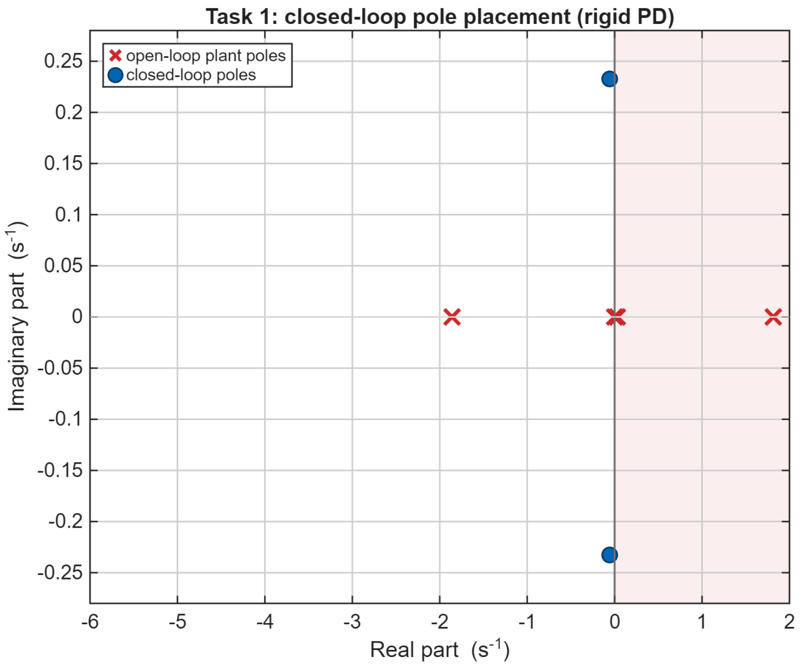

Figure 7: Task 1 — open-loop plant poles (crosses, one in the right half-plane) against the closed-loop poles (circles): the rigid PD stabilises the airframe.

2.3 Wind-gust response

The closed loop is excited by a deterministic "1-cosine" wind gust whose amplitude ($V_g = 6.4$ m/s) is taken from the reference dry-wind dispersion (`drywind.mat`, severe regime) at the current altitude, converted to a wind angle of attack $\alpha_w = v_w/V$ with a peak of 0.39°. The four key time histories requested by the assignment are shown in Figure 8: the pitch attitude peaks at 0.26°, the TVC deflection at 0.53°, and the lateral drift peaks at 2.3 m before the slow, lightly damped drift mode ($\zeta \approx 0.23$, $\omega_n \approx 0.24$ rad/s) returns it to trim. All four states decay back to zero within about 80 s — confirming the loop is asymptotically stable — and the actuator effort stays small.

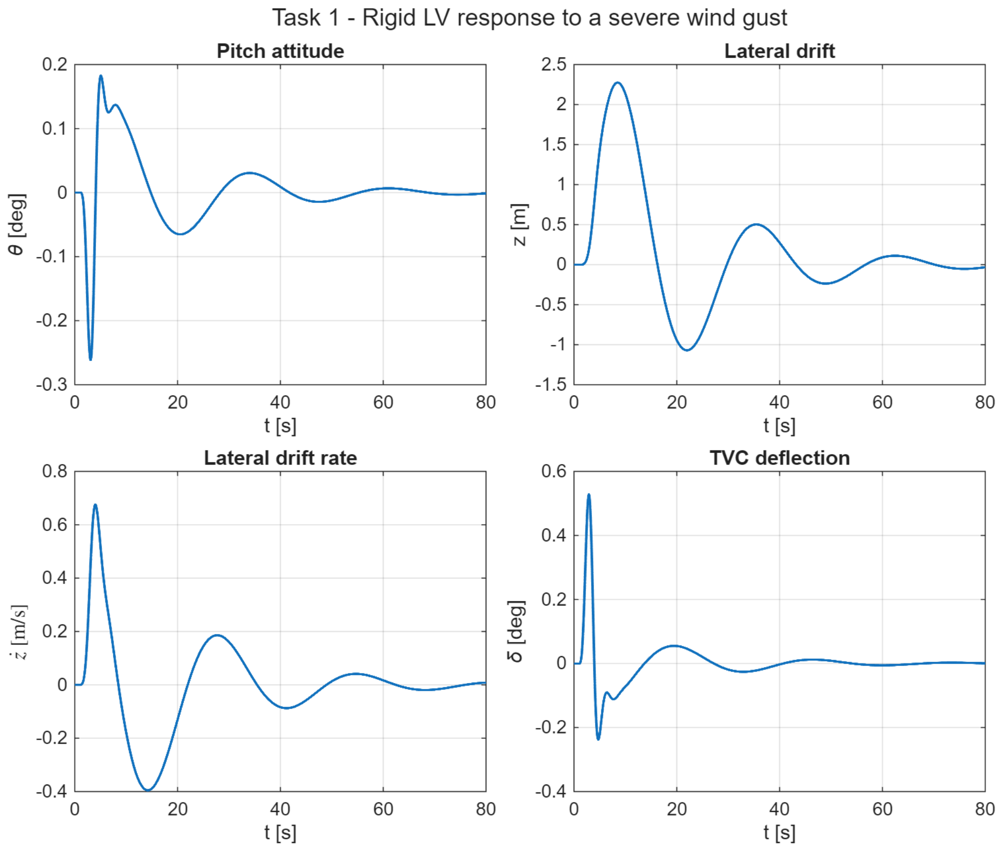

Figure 8: Task 1 — response to a severe wind gust: pitch attitude θ, lateral drift z, drift rate $\dot{z}$ and TVC deflection δ.

Since max-$\bar{q}$ is the load-critical point of the trajectory, the total angle of attack $\alpha = \theta + \dot{z}/V + \alpha_w$ and the structural-load indicator $\bar{q}\,\alpha$ are also monitored (Figure 9). The attitude loop pitches the vehicle slightly into the wind ($\theta \to -0.26°$ at the gust peak), so the peak total incidence, 0.26°, stays below the 0.39° wind contribution alone — a mild load-relieving action. With $\bar{q} = 81$ kPa the peak load indicator is $\bar{q}\,\alpha = 21$ kPa deg, the incidence remaining an order of magnitude below the few-degree load limit typical of a slender launcher, so at this gust level the drift-minimum–oriented gain choice does not conflict with the load-minimum requirement.

This drift-versus-load tension is the standard trade in launch-vehicle wind control [1, Sec. 3.1]. The characteristic equation of the rigid airframe in wind is a cubic with two oscillatory rigid-body roots plus one real drift (flight-path) root; the final-value theorem applied to a *sustained* step gust gives a steady drift rate $\dot{z}_{ss} = -V\,\alpha_w$, independent of the gains: with only weak drift feedback the vehicle does not null a persistent crosswind. The 1-cosine gust of Figure 8 is instead a transient pulse, so once it has passed the stable loop returns the drift to zero — but only through that slow, lightly damped mode, which is why the settling takes $\sim$80 s. Two limiting cases bracket the design. The **drift-minimum** condition (drift-channel coefficient $B_0 = 0$) holds the trajectory dispersion down. The **load-minimum** condition ($K_{P,\theta} = 0$, so the steady incidence $\alpha_{ss} = 0$) minimises $\bar{q}\,\alpha$, but pays for it with a slow unstable drift mode. The weak negative drift gains I used here are a small step toward the drift-minimum end of this trade.

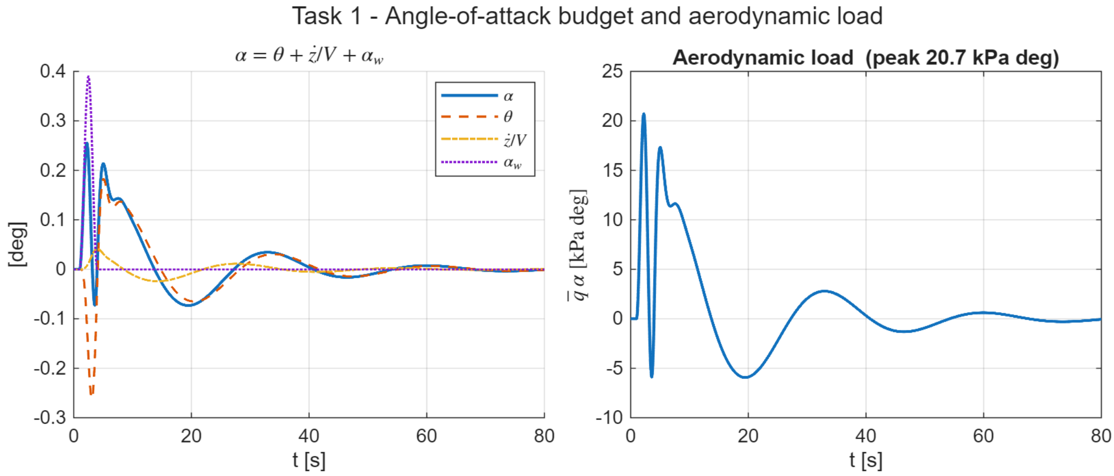

Figure 9: Task 1 — angle-of-attack budget $\alpha = \theta + \dot{z}/V + \alpha_w$ (left) and aerodynamic load indicator $\bar{q}\,\alpha$ (right) during the severe gust.

3 Task 2 — Full Launch Vehicle Model

Task 2 reintroduces the full dynamics of Eq. (1): the lightly damped first bending mode, the second-order TVC actuator of Eq. (4) with its 20 ms delay, and the INS measurement model of Eq. (3). I built the design up incrementally, as the assignment guidelines suggest, keeping the Task 1 PD gains fixed throughout. The guideline note says retuning is usually unnecessary, and that holds here.

3.1 Step B — uncompensated full model

Adding the TVC, delay and bending mode to the Task 1 controller leaves the low-frequency margins almost untouched, but the bending resonance shows up in the loop at $|L(\omega_{BM})| = +39$ dB. With a damping of only $\zeta_{BM} = 0.005$ this peak crosses the $-180°$ critical point and the closed loop goes **unstable** (a closed-loop pole at $\mathrm{Re}(s) = +5.6$). The red curve in Figure 10 shows the bending loop crossing into the unstable region.

3.2 Step C — bending filter trade study

The guideline filter of Eq. 4 admits two readings: as printed (negative numerator damping, a non-minimum-phase **lead-lag** that phase-stabilises the resonance by rotating it on the Nichols chart) or in its minimum-phase variant (a band-reject **notch** that gain-stabilises the resonance by driving it below 0 dB) [2, Lecture 18]. Instead of picking one up front, I compare four candidates with the same fixed PD gains (all built by `build_notch_filter.m`, Listing 3); Table 2 collects the resulting margins (rigid-body and minimum gain margin, phase margin, delay margin on top of the 20 ms already modelled, and the residual loop gain at the bending frequency).

- **C-LL — Eq. 4 lead-lag alone.** The filter is swept over the ranges suggested in the assignment ($\zeta_{x,N} \in [0.1, 0.3]$, $\zeta_{x,D} \in [0.4, 0.6]$, $\omega_x = \omega_{BM} \pm 4$ rad/s, 75 combinations): none stabilises the loop. The least-unstable combination ($\omega_x = 18.9$, $\zeta_{x,N} = 0.10$, $\zeta_{x,D} = 0.60$) still leaves a closed-loop pole at $\mathrm{Re}(s) = +0.16$ — the green curve in Figure 10 shows its bending lobe still pinned to the critical region. Phase rotation alone cannot rescue a +39 dB resonance here.
- **C-N — deep minimum-phase notch.** Gain stabilisation needs more than 39 dB of attenuation at ω_{BM}, which forces the numerator damping far below the suggested range: $\zeta_{x,N} = 0.002$, $\zeta_{x,D} = 0.7$ give a -51 dB null and drive $|L(\omega_{BM})|$ to -12 dB. The depth follows the closed-form attenuation of a second-order notch, $K_{\text{notch}} = 20\log_{10}(\zeta_{x,N}/\zeta_{x,D})$, so that $20\log_{10}(0.002/0.7) \approx -51$ dB at ω_{BM} [2, Lecture 18]; pulling a +39 dB peak below 0 dB requires $\zeta_{x,N}/\zeta_{x,D} \leq 10^{-39/20} \approx 0.011$, i.e. $\zeta_{x,N} \lesssim 0.008$ with $\zeta_{x,D} = 0.7$, which is why the sizing

has to leave the assignment's suggested $\zeta_{x,N} \in [0.1, 0.3]$ range. I keep the null narrow because the bending frequency sits only a decade above the rigid crossover ($\omega_{BM}/\omega_c \approx 8$): a wide band-reject section would leak phase lag into the conditionally stable low-frequency region.

- **C-T — notch triplet.** The notch-triplet recipe against bending-frequency uncertainty places three cascaded notches at $\{0.9, 1.0, 1.1\}\,\omega_{BM}$ [2, Lecture 18]. Sized to the same depth, however, the three wide ($\zeta_{x,D} = 0.7$) sections pile up about 30° of phase lag at the rigid crossover and the conditionally stable loop goes unstable at nominal ($|PM|$ collapses to 11.6°). This matches the course view: the triplet belongs to the higher, well-separated modes, while the first mode — too close to the control band — has to be handled either with an exact notch or by phase stabilisation.
- **C-NLL — notch + lead-lag.** Reading the guideline literally ("the Notch filter and possibly other filters"), the deep notch is cascaded with the Eq. 4 lead-lag, the partner swept over the same ranges: 31/75 combinations now stabilise the loop. The best one ($\omega_x = 22.9$, $\zeta_{x,N} = 0.10$, $\zeta_{x,D} = 0.40$, selected for maximum delay margin) is stable but with clearly thinner nominal margins ($|PM| = 9.2°$, min $|GM| = 0.7$ dB). Its payoff shows up in the robustness analysis below.

Table 2: Task 2 — bending-filter trade with fixed Task-1 PD gains.

Candidate	rigid $\lvert GM\rvert$ [dB]	min $\lvert GM\rvert$ [dB]	$\lvert PM\rvert$ [°]	DM [ms]	$\lvert L(\omega_{BM})\rvert$ [dB]	stable
B (no filter)	6.34	6.34	30.8	—	+38.8	no
C-LL (lead-lag)	6.58	0.61	9.0	—	+23.2	no
C-N (deep notch)	6.58	1.81	26.5	58.8	−12.1	**yes**
C-T (triplet)	7.07	1.48	11.6	—	−46.0	no
C-NLL (notch+LL)	6.72	0.73	9.2	21.9	−18.3	yes

I keep the **deep notch**: it gain-stabilises the resonance while preserving the rigid-body margins ($|GM| = 6.6$ dB, $|PM| = 26.5°$ — a few degrees below the 30° of the ideal-actuator loop of Task 1, the difference being the phase lag of the second-order TVC and the 20 ms transport delay), leaves a 58.8 ms delay margin, and needs no retuning of the PD gains, in line with the guideline note. The bending mode itself stays at its lightly damped but stable open-loop location. One caveat: the minimum gain margin of the stabilised loop, set by the bending shoulder, is 1.8 dB — fine for this exercise but well below the $\geq$ 12 dB of gain margin (and 45° of phase margin) usually required of gain-stabilised bending modes, as opposed to the 6 dB/30° rigid-body figure [1]. The tougher bending requirement comes from the larger modelling uncertainty and latency sensitivity at high frequency. Getting the deeper margin would cost a stiffer notch or a retuned loop gain.

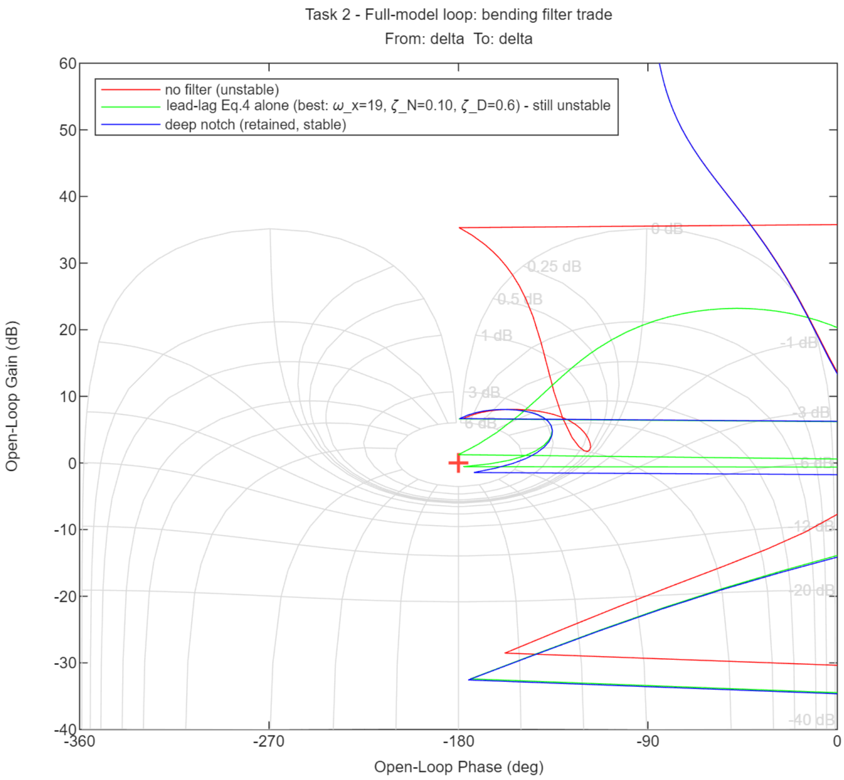

Figure 10: Task 2 — bending-filter trade on the Nichols chart (phase wrapped): without compensation the +39 dB resonance destabilises the loop (red); the best Eq.-4 lead-lag alone rotates the bending lobe but cannot clear the critical region (green); the deep notch gain-stabilises it and restores the rigid margins (blue).

The s-plane view of Figure 11 makes the mechanism explicit: without the notch the closed-loop bending poles are pushed into the right half-plane by the +39 dB resonance and the loop is unstable, whereas the deep notch drops the loop gain at ω_{BM} and returns them to their lightly damped, stable open-loop location just left of the imaginary axis. The notch gain-stabilises the mode; it does not add damping to it.

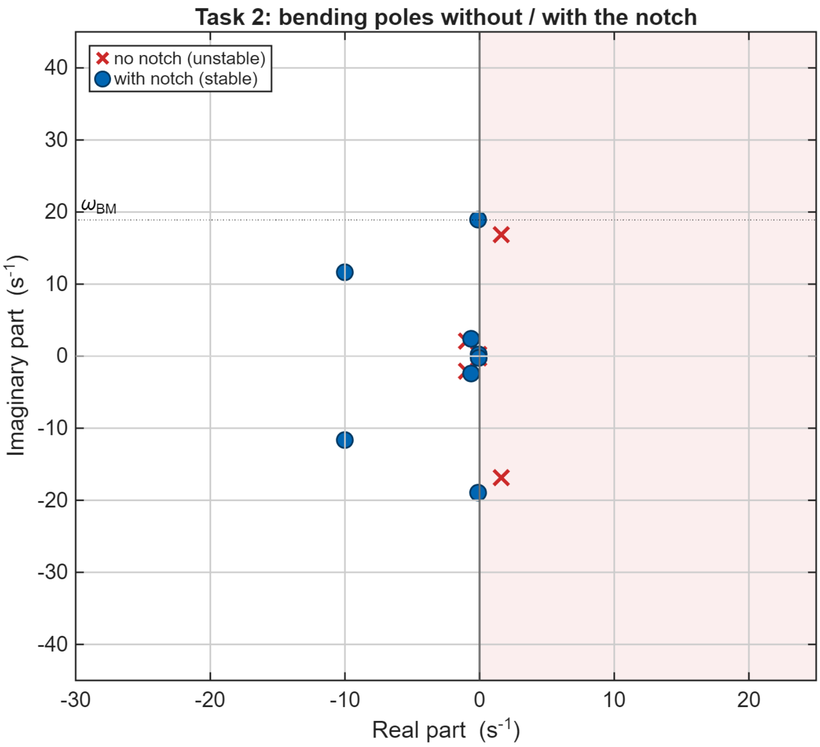

Figure 11: Task 2 — closed-loop bending poles without the notch (crosses, in the right half-plane) and with the notch (circles, returned to the stable neighbourhood of ω_{BM}).

The root locus of the full loop (Figure 12), from the built-in `rlocus`, tells the same story as a gain sweep: near ω_{BM} the bending branch skirts the imaginary axis while the notch keeps the closed-loop poles ($k = 1$) just inside the left half-plane. Its far branches race toward the right-half-plane zeros of the Padé delay approximation and are left off-view.

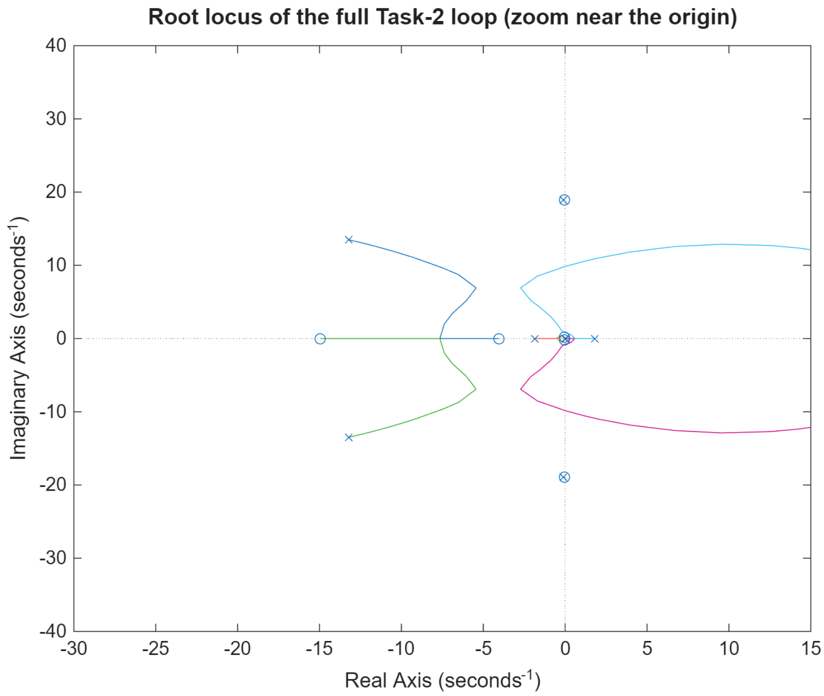

Figure 12: Task 2 — root locus of the full loop (flexible plant, TVC servo, Padé delay and notch), zoomed near the origin (×: open-loop poles; ○: open-loop zeros). The bending poles sit near $\omega_{BM} \approx 18.9$ rad/s and the design is stable (all closed-loop poles in the left half-plane); the delay's right-half-plane zeros lie far outside this window.

3.3 Sensitivity to the bending-frequency knowledge

The deep notch pays for its performance with a -51 dB null only 0.08 rad/s wide, i.e. it assumes ω_{BM} is known exactly — true in this exercise, where the LPV dataset provides it, but never true in practice, which is why the course notes prescribe the $\pm10\%$ triplet. Table 3 makes the trade explicit: each filter is held fixed while the true plant bending frequency is perturbed. The notch tolerates barely -5%; the notch+lead-lag combination, by phase-stabilising whatever leaks past the detuned notch, recovers nearly the full $\pm10\%$ band at the cost of the thin nominal margins seen above. In a flight design, where ω_{BM} drifts with propellant depletion, the robust variant (or a retuned phase-stabilised loop) would be mandatory; for this max-$\bar{q}$ snapshot I keep the exact notch.

Table 3: Task 2 — closed-loop stability with the filters fixed and the true bending frequency perturbed by up to $\pm10\%$.

Candidate	$0.90\,\omega_{BM}$	$0.95\,\omega_{BM}$	ω_{BM}	$1.05\,\omega_{BM}$	$1.10\,\omega_{BM}$
C-N — deep notch	no	yes	yes	no	no
C-T — notch triplet	no	no	no	no	no
C-NLL — notch+lead-lag	no	yes	yes	yes	yes

3.4 Validation and comparison with the rigid model

The gust response of the stabilised full model is shown in Figure 13; all four assignment time histories remain well damped, with peaks $\theta = 0.132°$, $z = 2.37$ m, $\delta = 0.39°$ and a load indicator $\bar{q}\,\alpha = 24.9$ kPa deg, just below the rigid case. Because the notch is localised at the bending frequency and the TVC bandwidth (70 rad/s) is well above the control crossover, the full-model response is almost indistinguishable from the rigid one (Figure 14): the added dynamics affect stability robustness (the bending and delay margins) far more than nominal performance.

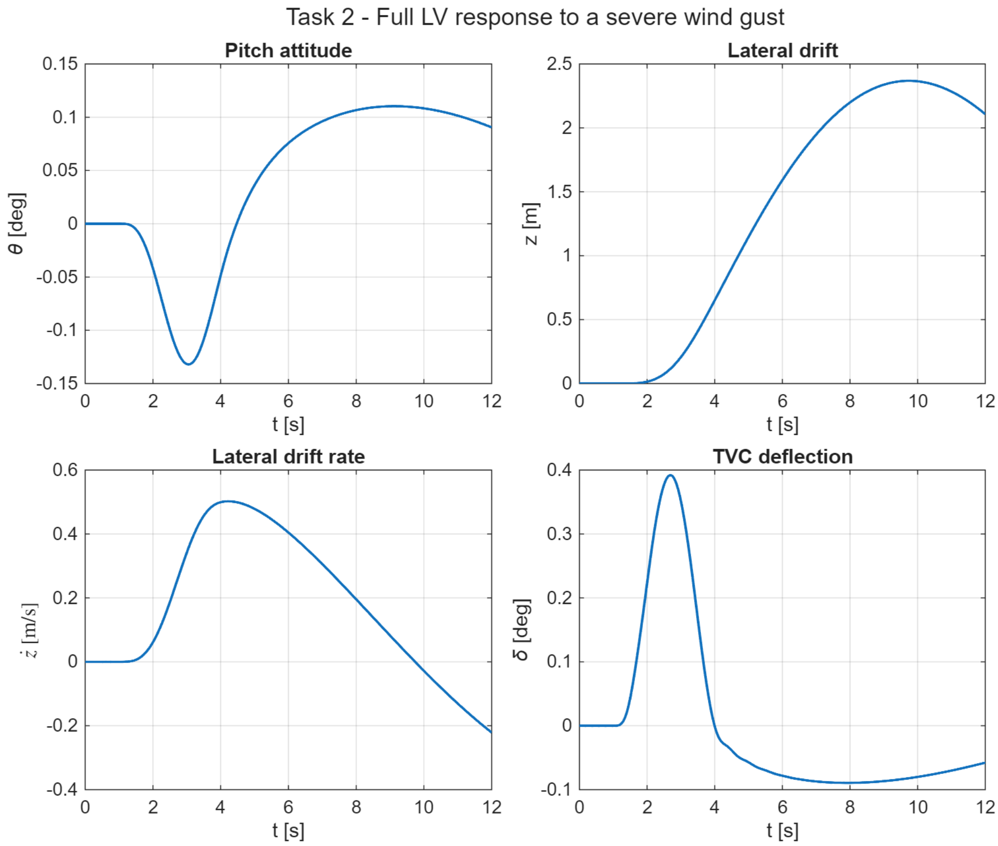

Figure 13: Task 2 — full-model response to the severe wind gust: pitch attitude θ, lateral drift z, drift rate $\dot{z}$ and TVC deflection δ.

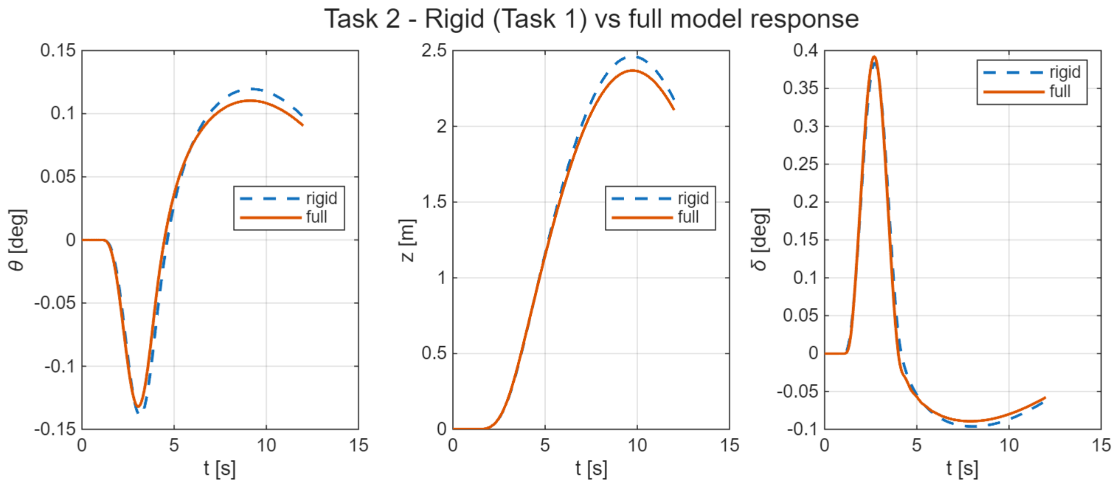

Figure 14: Task 2 — rigid (Task 1) versus full-model gust response: pitch attitude, lateral drift and TVC deflection are nearly coincident.

3.5 A note on the tail-wags-dog effect

A second flexible effect bounds what any filter can achieve. The gimballed nozzle has its own inertia, so its angular acceleration reacts back on the airframe. Following the course treatment [2, Lecture 20], the oscillating nozzle exerts on the vehicle a lateral reaction force and a moment

$$R = -\,m_E \ell_E\, \ddot{\delta}, \qquad M_{TWD} = \left(I_E + m_E \ell_E^2\right) \ddot{\delta}, \tag{11}$$

proportional to the gimbal acceleration $\ddot{\delta}$ (m_E nozzle mass, ℓ_E pivot-to-nozzle-CG distance, I_E nozzle inertia). This **tail-wags-dog** (TWD) coupling feeds the second derivative of the control back into the plant; it adds phase lag and an effective transmission zero that caps the achievable closed-loop bandwidth. The $-\phi_{1,TVC} T_c$ entry in the bending-forcing column of Eq. (1) is the flexible-mode analogue of this nozzle-reaction term. My model treats the actuator as a stiff second-order block (Eq. (4)) and so neglects the nozzle back-reaction — the dog-wags-tail feedback, which would add the gimbal angle as an extra state. A higher-fidelity design would keep it, one of the extensions noted in the conclusions.

4 Task 3 — Robustness to Parametric Uncertainty

The aerodynamic moment $\mu_\alpha = A_6$ and the control effectiveness $\mu_c = K_1$ are treated as uncertain. The controller designed in Task 2 (PD gains plus bending notch) is held **fixed** and I assess its robustness over the four corner cases of the uncertainty box, i.e. the vertices V1–V4 obtained by independently varying μ_α and μ_c by ±30%. Since each vertex perturbs both parameters at once, I also evaluate the four one-at-a-time variations S1–S4 (±30% on a single parameter): they sit on the edge midpoints of the box rather than on its corners, but they attribute the observed effects to each parameter.

4.1 Frequency- and time-domain analysis

Table 4 collects, for every case, the rigid-body (low-frequency) gain margin, the smallest gain margin over the whole loop (governed by the bending shoulder), the phase margin, the delay margin, and the peak pitch and lateral-drift excursions to the same severe gust. All vertices stay stable: the fixed Task-2 controller tolerates the entire ±30% uncertainty box without re-design.

Table 4: Task 3 — margins and gust-response peaks: box vertices (V1–V4, the assignment corner cases) and one-at-a-time sensitivities (S1–S4).

Case	μ_α	μ_c	rigid $\|GM\|$ [dB]	min $\|GM\|$ [dB]	$\|PM\|$ [°]	DM [ms]	peak θ [°]	peak z [m]
Nominal	1.00	1.00	6.58	1.81	26.5	58.8	0.132	2.37
V1	0.70	0.70	6.42	2.81	31.3	134.9	0.136	2.38
V2	0.70	1.30	11.68	0.88	11.3	19.2	0.061	1.67
V3	1.30	0.70	**1.53**	1.53	**9.1**	156.4	**0.353**	**4.20**
V4	1.30	1.30	6.66	0.93	12.3	21.3	0.130	2.36
S1	0.70	1.00	9.44	1.79	25.7	56.1	0.085	1.91
S2	1.30	1.00	4.47	1.83	22.1	61.8	0.187	2.88
S3	1.00	0.70	3.59	2.82	18.6	144.6	0.219	3.16
S4	1.00	1.30	8.79	0.91	11.8	20.2	0.093	2.01

The Nichols overlay over the nominal case and the four vertices (Figure 15) shows the curves clustering at low frequency and fanning out near the bending frequency; the time-domain overlay (Figure 16) shows the corresponding spread of the gust response.

4.2 Discussion

The sensitivities S1–S4 attribute the effects: μ_α moves the unstable airframe pole (±30% shifts the rigid margin by roughly ∓2–3 dB and scales the gust excursions), while μ_c scales the whole loop gain (+30% improves the rigid margin but pushes the gain-stabilised bending shoulder 2.3 dB closer to 0 dB — the leading-order $20 \log_{10} 1.3 = 2.28$ dB loop-gain shift, only approximate because K_1 also scales the bending-forcing column of Eq. (1) and not merely the loop gain — and halves the delay margin). The vertices then combine these mechanisms:

- **V3** ($\mu_\alpha\uparrow$, $\mu_c\downarrow$) is the worst case for the rigid loop and for performance, because the two effects compound: the airframe is more unstable and the authority to fight it is reduced at the same time. The rigid gain margin collapses to 1.5 dB with $|PM| = 9.1°$, and the gust excursions grow to 0.35° in pitch and 4.2 m of drift — roughly 60% beyond the worst one-at-a-time case (S3). This is why I test the actual vertices and not only the single-parameter variations.
- **V2 and V4** ($\mu_c\uparrow$) are the worst cases for the bending channel: the loop-gain increase squeezes the minimum gain margin to 0.9 dB and the delay margin to ~20 ms. The binding constraint in these corners is the flexible side of the loop, not the rigid one.
- **V1** ($\mu_\alpha\downarrow$, $\mu_c\downarrow$) is benign on both counts: the two reductions nearly cancel in the rigid loop and relax the bending shoulder.

So performance degradation is driven by μ_α and reduced by μ_c, while the bending channel loses stability margin as μ_c grows, and the worst-case combination V3 leaves single-digit margins. Because the margins are non-monotonic in μ_c (it helps the rigid loop while hurting the bending shoulder), the global worst case over the ±30% box need not sit on a vertex: the four corners and the one-at-a-time sensitivities are only a screening, and a denser grid or Monte-Carlo sweep would be needed to certify an interior worst case. The fixed classical design survives the whole box, but a requirement of, say, ≥ 3 dB and $\geq 10°$ in every corner would already call for either gain scheduling on $\bar{q}$ or a robust-synthesis retune, the standard routes beyond a single fixed classical design. That is the practical takeaway here.

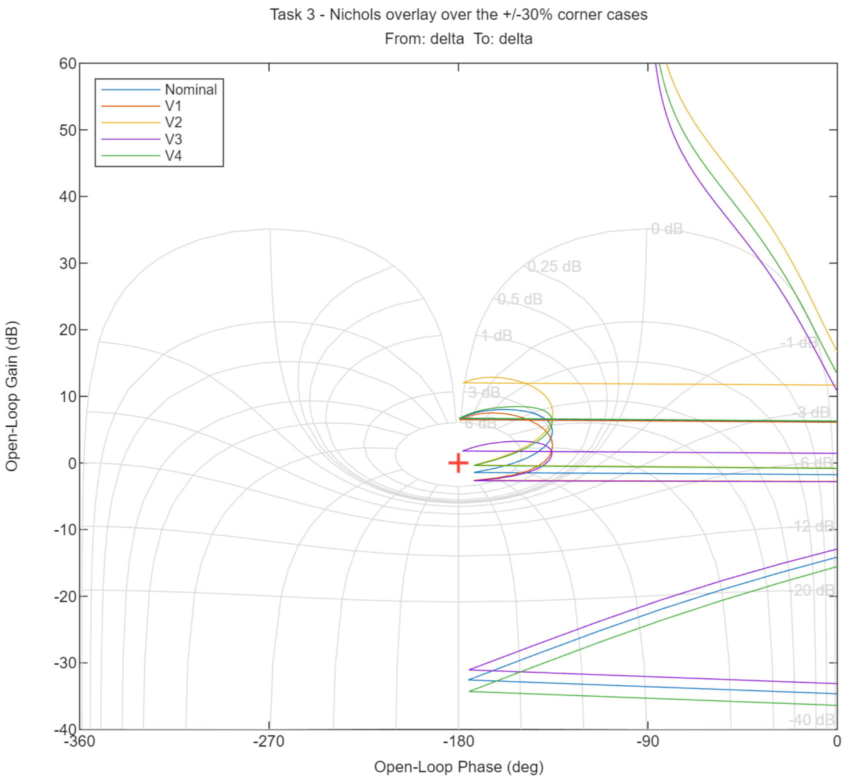

Figure 15: Task 3 — Nichols overlay over the nominal case and the four uncertainty-box vertices.

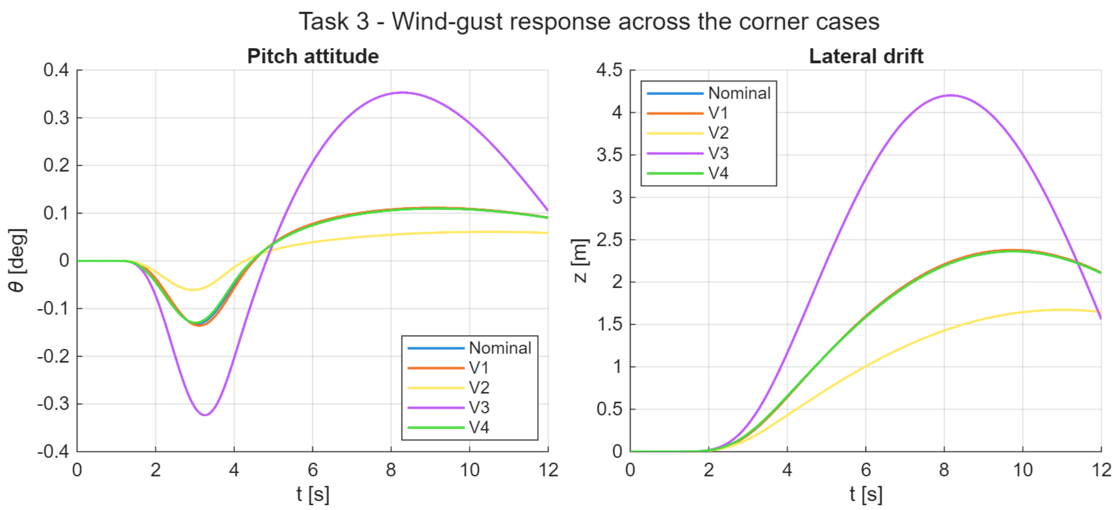

Figure 16: Task 3 — wind-gust response over the nominal case and the four vertices: pitch attitude (left) and lateral drift (right).

Figure 17 closes the frequency-domain view in the s-plane: the closed-loop poles for the nominal case and the four vertices all stay in the left half-plane, so the fixed classical design is robustly stable over the whole $\pm 30\%$ box. The lightly damped bending poles near ω_{BM} move the most and sit closest to the imaginary axis, consistent with the thin bending-shoulder margin identified above.

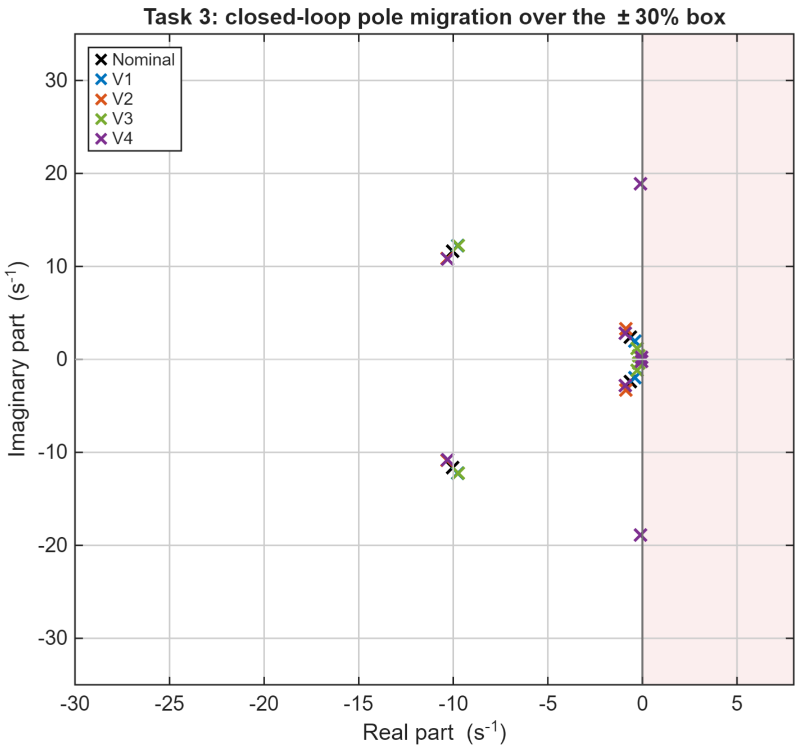

Figure 17: Task 3 — closed-loop pole migration over the nominal case and the four ±30% vertices (zoom near the imaginary axis); all poles remain in the left half-plane.

5 Conclusions

I designed and validated a classical attitude control system for an aerodynamically unstable launch vehicle at the max-$\bar{q}$ condition, working entirely in the frequency domain.

- **Task 1.** A proportional–derivative attitude law with weak negative drift feedback stabilises the rigid airframe (open-loop pole at +1.84 rad/s) and meets the design targets $|GM_a| \approx 6$ dB, $PM \approx 30°$ — the aerodynamic gain margin and rigid-body phase margin, read on the full conditionally-stable loop and classified by frequency band.
- **Task 2.** Introducing the TVC actuator, the transport delay and the lightly damped bending mode destabilises the loop through a +39 dB resonance. A four-way filter trade showed that the Eq.-4 lead-lag alone never stabilises the loop within the suggested parameter ranges (0/75 combinations), and that a notch triplet sacrifices too much phase at low frequency for a conditionally stable loop; a deep gain-stabilising notch centred on ω_{BM} restores stability while preserving the rigid margins, with negligible change to the nominal gust response. I quantified how much it leans on knowing the bending frequency exactly (±5% detuning is fatal) and found a robust notch+lead-lag variant that tolerates nearly ±10% at the price of thinner nominal margins.
- **Task 3.** The fixed Task 2 controller is robust to ±30% independent variations of μ_α and μ_c: all four vertices of the uncertainty box remain stable. Uncertainty on μ_α erodes the rigid margin and the gust performance, uncertainty on μ_c erodes the bending and delay margins, and the worst vertex (high μ_α with low μ_c) compounds them, leaving only 1.5 dB and 9° — the quantitative argument for gain scheduling in a flight design.

Limitations and extensions. Four modelling choices bound the scope of this design. (i) The pure proportional–derivative law leaves a steady-state attitude error to a constant gimbal misalignment, which a proportional–integral–derivative extension would null. (ii) The stiff second-order TVC model neglects the nozzle back-reaction (the dog-wags-tail feedback) and the hydraulic load-resonance of the actuator; retaining the gimbal angle as a state, together with the tail-wags-dog reaction of Eq. (11) [2, Lecture 20], is the natural next fidelity step. (iii) Sensor placement relative to the bending-mode nodes matters as much as the filter — an aft gyro station flips the sign of the modal slope and can phase-stabilise a mode that would otherwise need a notch [1, Sec. 3.3.2.2] — whereas here the INS station is fixed and only the filter is traded. (iv) Propellant sloshing is omitted; it would add low-frequency, lightly damped modes coupling into the rigid loop and the drift channel.

The design shows the textbook trade-off of launch-vehicle attitude control [2, Lecture 18]: enough gain to stabilise the unstable airframe, but not so much that it excites the structural modes, and the Nichols chart lets me see both constraints on one plot. A Simulink block-diagram mirror of the closed loop reproduces the script results and is documented in the accompanying `models/SIMULINK_GUIDE.md`.

A Implementation: Key Code Excerpts

The full MATLAB sources live in the `HM3/` folder of the repository: `main_task1.m`, `main_task2.m` and `main_task3.m` drive the three tasks, and the helper functions (`load_hw3_params`, `build_plant_rigid`, `build_plant_full`, `build_tvc`, `build_notch_filter`, `design_controller`, `assemble_loop`, `simulate_gust_response`) are shared across them. This appendix collects the three routines that set up the control problem: the rigid plant, the loop closure the Nichols analysis is drawn from, and the bending notch.

Listing 1 builds the rigid pitch-plane state-space model at the max-$\bar{q}$ condition by dropping the bending states from the assignment dynamics; the unstable airframe pole sits at $+\sqrt{A_6}$.

```
function G = build_plant_rigid(p)
% Rigid pitch-plane LV plant at max-qbar (Task 1, Eq. 1 minus bending).
%   INPUT
%     p - param struct (load_hw3_params)
%   OUTPUT
%     G - ss, 4 states [z zdot theta thetadot], in [delta alpha_w],
%         out [theta_m thetadot_m z_m zdot_m theta z zdot]
%   First 4 outputs feed the controller (= true states here, no bending);
%   last 3 are plotting signals.

A = [0   1        0          0;
     0   p.a1     p.a1*p.V+p.a4   0;
     0   0        0          1;
     0   p.A6/p.V p.A6       0];

Bd = [0; p.a3; 0; p.K1];           % delta column
Bw = [0; -p.a1*p.V; 0; -p.A6];     % alpha_w column

% Outputs: [theta_m, thetadot_m, z_m, zdot_m, theta, z, zdot]
%           state index:   z=1  zdot=2  theta=3  thetadot=4
C = [0 0 1 0;     % theta_m    = theta
     0 0 0 1;     % thetadot_m= thetadot
     1 0 0 0;     % z_m        = z
     0 1 0 0;     % zdot_m     = zdot
     0 0 1 0;     % theta
     1 0 0 0;     % z
     0 1 0 0];    % zdot
D = zeros(7, 2);

G = ss(A, [Bd Bw], C, D);
G.StateName  = {'z','zdot','theta','thetadot'};
G.InputName  = {'delta','alpha_w'};
G.OutputName = {'theta_m','thetadot_m','z_m','zdot_m','theta','z','zdot'};
G.Name = 'Plant_Rigid';
end
```

Listing 1: Rigid pitch-plane LV plant (`build_plant_rigid.m`).

Listing 2 closes the PD attitude and drift loop around the plant, optionally inserting the TVC, delay and notch chain through `Wact`, and returns the SISO open-loop transfer broken at the TVC deflection. This open-loop transfer is what the Nichols charts in

Tasks 1–3 are built from.

```

% --- open loop at the break point 'delta' (1+L convention) ---
L = getLoopTransfer(T, 'delta', -1);
L = minreal(tf(L), 1e-6);

info = struct('Kc', Kc, 'Wact', Wa);
end
```

Listing 2: Loop closure and open-loop transfer for the Nichols analysis (`assemble_loop.m`).

Listing 3 is the second-order lead-lag / notch section used in Task 2 to stabilize the lightly damped first bending mode.

```
function Hx = build_notch_filter(wx, zN, zD, numSign)
% Lead-lag / notch section for bending stabilisation (Eq. 4).
%   Hx(s) = (s^2 + sgn*2*zN*wx*s + wx^2) / (s^2 + 2*zD*wx*s + wx^2)
%   INPUT
%     wx      - centre frequency [rad/s] (~ wBM +/- 4)
%     zN      - numerator damping (guideline 0.1-0.3)
%     zD      - denominator damping (guideline 0.4-0.6)
%     numSign - -1 (default, Eq. 4 as printed): RHP zeros, non-min-phase
%               phase-shaper; +1: min-phase symmetric notch
%   OUTPUT
%     Hx - tf

arguments
    wx (1,1) {mustBeNumeric, mustBeReal, mustBePositive}
    zN (1,1) {mustBeNumeric, mustBeReal, mustBeNonnegative}
    zD (1,1) {mustBeNumeric, mustBeReal, mustBePositive}
    numSign (1,1) {mustBeMember(numSign, [-1, 1])} = -1
end

num = [1, numSign*2*zN*wx, wx^2];
den = [1,         2*zD*wx, wx^2];

Hx = tf(num, den);
Hx.Name = 'Notch_Hx';
end
```

Listing 3: Lead-lag / notch filter for bending stabilization (`build_notch_filter.m`).

References

[1] A. L. Greensite. Analysis and design of space vehicle flight control systems. volume vii — attitude control during launch. Technical Report CR-826, NASA, Washington, DC, July 1967.

[2] A. Zavoli. Dynamics and control of launch vehicles — course lecture notes. Graduate course, Sapienza University of Rome, 2026. Academic Year 2025/26.

[3] Timothy M. Barrows and Jeb S. Orr. *Dynamics and Simulation of Flexible Rockets*. Academic Press, 2020.